

Khulna University of Engineering & Technology

Department of Computer Science and Engineering

CSE 4112: Machine Learning Laboratory

DATASET REPORT

A Speaker-Attributed Sentiment Dataset for Bangla Financial Television Talk Shows

Group No.	1	Dataset Version	v1.0 (15 September 2026)
Students	Nafiz Ahmed (2107001) • Tawhidul Hasan (2107004) • Sarwad Hasan Siddiqui (2107006) • Saif Ahmed Shejan (2107009) • Maruf Shafiq (2107010) • Dewan Salman Rahman Zisan (2107015)	Repository	https://github.com/mayer-doa-coder/bangla-financial-talkshow-sentiment

(a) Framework illustrating the practical application of the data

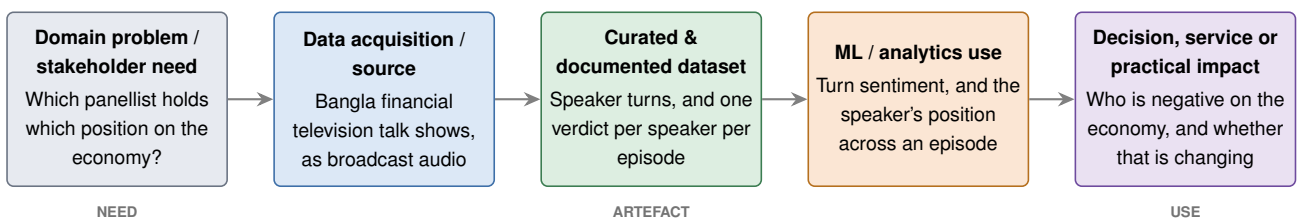


Figure 1(a). Practical application framework. The intended users are financial-media monitors, macroeconomic policy analysts and Bangla NLP researchers. The practical outcome is a stance signal resolved to the individual speaker, which a document-level sentiment label cannot produce because a panel programme contains people who disagree with each other.

(b) Technical pipeline of the ML method using the dataset

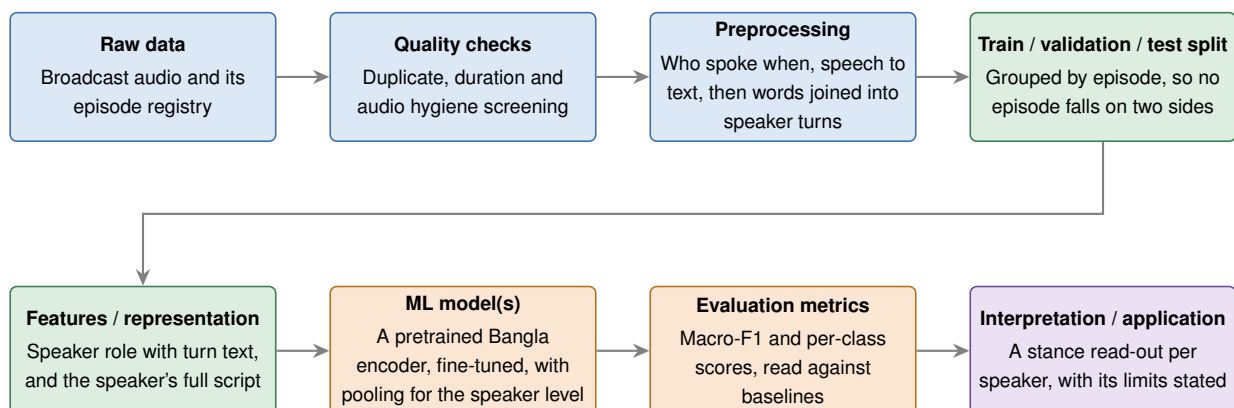


Figure 1(b). Technical pipeline followed in the course project, shown at stage level. Splitting is grouped by episode rather than by row, so turns from one panel cannot fall on both sides of the split, and the duplicate episode is removed before partitioning. Section 6 gives the same pipeline with the models, settings and counts actually used.

REPORT PREPARATION AND SUBMISSION FORMAT

This is a course-specific dataset report template. It combines the reporting logic of *Data in Brief* with common data-descriptor elements used by *Scientific Data* and *MDPI Data*, and adds an ML-readiness/baseline section for CSE 4112.

Paper & margins	A4 (210 × 297 mm); approximately 20 mm margins.
Font	Times New Roman, 11 pt body; Arial for headings/figures. Keep formatting consistent.
Spacing	1.15 line spacing; 6 pt after body paragraphs. Single spacing is acceptable in tables/captions.
Recommended length	About 8–12 pages after Page 1, excluding appendices. Quality and completeness are more important than page count.
Figures & tables	Number consecutively; cite every figure/table in the text; captions must be self-explanatory.
References	Numbered citation style in square brackets. Prefer dataset papers, data-collection sources, instruments/software, and relevant ML references.
Submission files	One report (DOCX/PDF as instructed), dataset/repository link, README, data dictionary, and code/notebook link where applicable.

1. Dataset Article Information

Dataset/report title	A Speaker-Attributed Sentiment Dataset for Bangla Financial Television Talk Shows
Group and students	Group 1; six students: Nafiz Ahmed (2107001), Tawhidul Hasan (2107004), Sarwad Hasan Siddiqui (2107006), Saif Ahmed Shejan (2107009), Maruf Shafiq (2107010), Dewan Salman Rahman Zisan (2107015)
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Corresponding student	Tawhidul Hasan (2107004), hasan2107004@stud.kuet.ac.bd
Keywords	Bangla NLP; speaker-level sentiment; financial talk shows; speaker diarization; automatic speech recognition; LLM-as-judge; BanglaBERT; low-resource corpus
Dataset version	v1.0; released 15 September 2026
Repository / persistent ID	Git repository: https://github.com/mayer-doa-coder/bangla-financial-talkshow-sentiment . Access is currently restricted to the course instructor and the authors; no DOI is minted yet. Source broadcast audio and the fine-tuned checkpoint are excluded; see Section 13 and the repository .gitignore.
License	CC BY 4.0 for the derived annotations, transcripts and timings <i>only</i> . Source broadcast audio is third-party copyrighted and is not covered by this licence and not redistributed (Section 13).
Related article / source	None

2. Abstract

Write 150–300 words. Describe why the dataset was created, how data were collected/compiled, what the dataset contains, its scale/formats, quality-control approach, and how it may be reused. Do not turn the abstract into a results or conclusion section.

Bangla financial television talk shows carry the evaluative positions of economists, bankers, industrialists and regulators, but existing Bangla sentiment resources are built from social-media text and score whole documents rather than individual speakers. This dataset was created to make *who holds which position* machine-readable. Forty-five unique episodes (approximately 28.6 hours) from Bangladeshi business talk shows were processed through a reproducible pipeline: audio hygiene and voice-activity detection, `pyannote` speaker diarization, Whisper-medium Bangla ASR, and word-level fusion of the two streams. Fusion output was merged from fixed

ASR decode windows back into 2,414 coherent speaker turns across 212 speaker–episode units. Because no human sentiment labels existed for this genre, labels were produced by a large-language-model judge operating under a published rubric that separates evaluative stance from emotion, treats a host’s pointed question as elicitation rather than assertion, and allows an explicit *illegible* class for turns the ASR destroyed. The release contains 2,414 turns, 150 judged speaker units, 1,811 turn labels and 1,630 supervised turns partitioned episode-disjointly into 1,009/297/324. A BanglaBERT baseline reaches a test turn-level macro-F1 of 0.670 ($\kappa = 0.496$) against a majority-class floor of 0.236 and a TF-IDF reference of 0.529, and a hierarchical speaker model scores 0.530 macro-F1 in grouped cross-validation over all 147 speaker units, demonstrating that the corpus supports a valid supervised workflow at both levels. **All reported scores measure agreement with the judge, not with human ground truth;** word error rate, diarization error rate and annotator agreement remain unmeasured and are documented as such.

3. Dataset Specifications Table

Item	Required information
Subject / domain	Computer science and natural language processing; applied economics / financial media analysis
Specific subject area	Speaker-attributed sentiment and stance analysis of Bangla-language broadcast financial discourse
Type of data	Multimodal: raw broadcast audio (not redistributed), processed WAV, ASR transcripts, diarization segments, fused word–speaker alignments, speaker turns, and sentiment labels. Both raw and processed components are released.
File formats	MP3/MPEG (source), WAV 16 kHz mono (processed), JSON (ASR, diarization, fusion), RTTM (diarization, standard), JSONL (turns, speakers, labels, splits), CSV (registry, lexicons), Jupyter .ipynb
Unit of observation	One speaker turn : a maximal run of consecutive ASR decode windows attributed to the same speaker within one episode. A secondary unit is the speaker–episode unit (all turns by one speaker in one episode).
Data collection / source	Secondary compilation: publicly broadcast television talk-show recordings collected by six students, then processed by an automatic pipeline. Sentiment labels are primary annotations generated by an LLM judge.
Instrument / software	pyannotate/speaker-diarization-community-1 (rev 3533c8cf) with wespeaker-voxceleb-resnet34-LM embeddings; bengaliAI/tugstugi_bengali-ai-asr_whisper-medium (rev da605cc1) via CTranslate2; csebuetnlp/banglabert (rev 9ce791f3); Claude Opus 5 judge; PyTorch 2.12, transformers, scikit-learn
Data source location	Bangladesh; Bangla-language national business/economics television programmes (ATN Bangla, RTV Business, Desh TV, GTV, Kaler Kantho, Star News, Dhaka Stream and others)
Collection period	Original broadcast dates are not recorded in the source postings and are therefore not claimed; <code>registry.csv</code> carries channel, title and duration but no air date. Episodes were retrieved from public platform postings and the pipeline runs were executed August–September 2026 (second build).
Dataset size	45 unique episodes (≈ 28.6 h audio); 5,327 ASR decode windows; 2,414 speaker turns ($\approx 219,000$ words); 212 speaker–episode units; 150 judged units; 1,811 turn labels; 1,630 supervised turns
Target / labels	Turn level: {negative, neutral, positive} (plus <i>illegible</i> , excluded from supervision). Speaker level: {negative, mixed, neutral, positive} plus a continuous <code>stance_score</code> $\in [-1, 1]$.

Item	Required information
Accessibility	Git repository (restricted at submission): https://github.com/mayer-doa-coder/bangla-financial-talkshow-sentiment . It carries the derived annotations, transcripts, timings, diarization output, the build and baseline code and this report; the source broadcast audio is not redistributed.
License / reuse terms	CC BY 4.0 for the derived annotations; source broadcast audio is third-party copyrighted and is not redistributed; see Section 13.

4. Value of the Data

Provide 3–6 concise bullet points. Explain why the data are useful, who can reuse them, what research/engineering questions they can support, and what makes the dataset practically relevant. Avoid unsupported claims of novelty or superiority.

- The data attach sentiment to **individual speakers** rather than to whole episodes. In a panel discussion the host, the economist and the industry representative frequently disagree; a document-level label averages that disagreement away and discards exactly the information a financial sentiment signal needs.
- The corpus is a rare Bangla resource in the **spoken, broadcast, financial** domain. Published Bangla sentiment datasets are overwhelmingly written social-media text, so this release supports questions about domain shift from written to ASR-transcribed speech.
- It supports several ML tasks directly: turn-level sentiment classification, speaker-level stance aggregation, role classification (host vs. guest), hierarchical document modelling over speaker scripts, and robustness studies on noisy ASR output.
- Every intermediate artefact is preserved: diarization RTTM, per-word ASR probabilities, fusion alignments, and judge rationales with evidence quotes. The corpus can therefore be re-derived under a different diarizer, a different ASR model or a different labelling rule without re-collecting audio.
- The dataset is extensible along two documented axes: the audio is retained, so acoustic and multimodal models can be added; and the existing 52-entry financial-target lexicon allows extension to speaker $\times$ target aspect-based sentiment.
- The quality defects are documented rather than hidden (Sections 9 and 12), which makes the corpus usable as a teaching example of leakage control and honest validation reporting.

5. Background and Practical Motivation

Explain the real-world problem and the motivation for creating the dataset. Briefly position it relative to existing datasets or data-collection practices. Refer back to the practical-application framework on Page 1. Recommended: 200–400 words.

Bangladesh’s recent macroeconomic period, marked by banking-sector stress, foreign-reserve pressure and sustained inflation, was debated nightly on television business programmes. These broadcasts are the most concentrated public record of expert opinion on the economy, yet they are audio, in Bangla, and unindexed. Analysts who want to know whether professional sentiment is turning must watch them.

The gap is twofold. First, Bangla sentiment resources (BLP shared tasks, BanglaBERT’s BLUB benchmark, BAN-ABSA and BABSAs) are built from written social-media or review text; none covers broadcast financial speech. Second, and more fundamentally, almost all conversational sentiment resources label *utterances* or *documents*, not *speakers*. As Figure 1(a) shows, the practical output required by the stakeholders (media

monitors, policy analysts and researchers) is a statement of the form “this economist is negative on the banking sector,” which a document-level label cannot express.

This motivated a dataset whose unit of analysis is the speaker turn and whose secondary unit is the speaker-episode. Producing it required solving an attribution problem before any sentiment problem: television panels overlap, interrupt and hand over rapidly, so speaker identity must be recovered from the audio and then aligned to ASR words. The pipeline in Figure 2 is the result. A final constraint shaped the labelling strategy: the collected material carried **zero human sentiment labels**, and the only pre-existing annotations in the project (180 target-level pairs) were single-annotator and self-declared as provisional. LLM-as-judge labelling was therefore adopted as the principal supervision source, with its limitations reported explicitly rather than absorbed silently.

6. Data Acquisition / Experimental Design, Materials and Methods

Describe data generation/collection in enough detail that another group could reproduce the process. Distinguish primary data from externally sourced/secondary data.

Figure 2 sets out the processing chain as it was actually run. Every stage writes a durable artefact to disk, so any step can be inspected or repeated without re-running the ones before it.

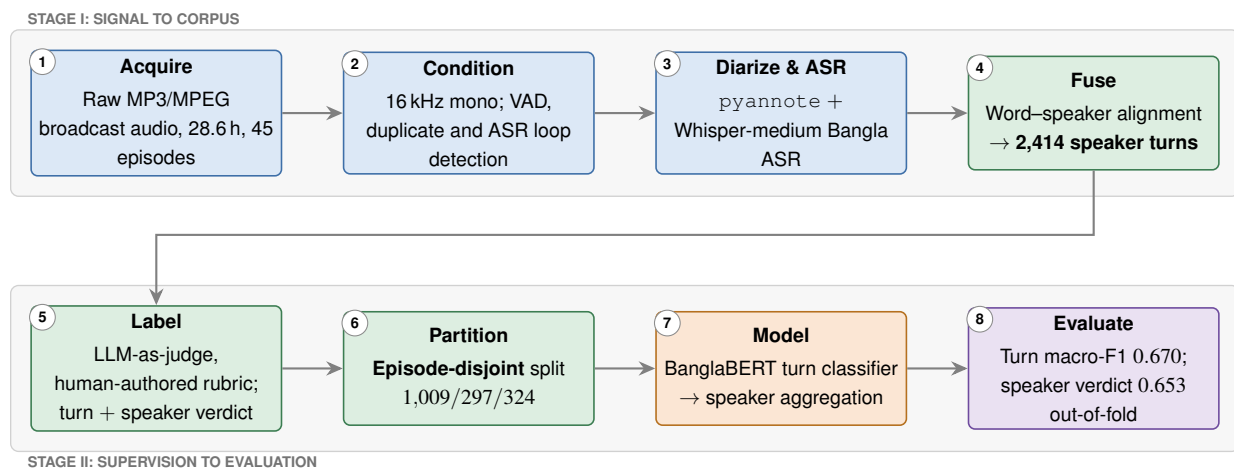


Figure 2. The same pipeline as Figure 1(b), now with the models, settings and counts actually used. Splitting is grouped by episode, never by row, so that turns from one panel cannot appear on both sides of the split; the deduplicated episode is removed before partitioning. All 45 episodes carry trusted `pyannote` diarization in this build.

6.1 Data source and sampling strategy

The source frame is publicly broadcast Bangla-language television programmes devoted to business and economics. Each of the six group members independently collected five to six episodes into a roll-numbered folder (2107001, 2107004, 2107006, 2107009, 2107010, 2107015), giving 56 registry entries and 46 processed episodes. Inclusion required a multi-speaker panel or interview format, a predominantly economic topic and intelligible audio. Two exclusions were applied after profiling: one episode (2107010::ep006) was found to be byte-identical in transcript content to 2107004::ep006 and was removed by transcript hash, leaving **45 unique episodes**; and one episode (2107015::ep002) proved to be an interview on law and order rather than economics, and is retained but flagged out-of-domain.

6.2 Collection setting, period and location

Episodes are studio-recorded broadcasts from Bangladeshi national channels, retrieved as MP3/MPEG audio. Durations range from 1,411 s to 3,614 s (mean $\approx 2,420$ s), totalling approximately 17.5 hours. Pipeline execution took place on Google Colab (CUDA device) in August 2026 with fixed seed 17; the speaker-sentiment stage was executed locally and on Colab with seed 7.

6.3 Instruments, hardware, software and versions

Stage	Component, version and settings
Audio hygiene	Resample to 16 kHz mono WAV; loudness and clipping report written to <code>processed/hygiene_report.json</code>
Diarization	pyannote/speaker-diarization-community-1, revision 3533c8cf8e369892e6b79ff1bf80f7b0286a54ee, with wespeaker-voxceleb-resnet34-LM embeddings; min_duration_off 0.1 s, min_cluster_size 20, min_segment_sec 0.75, min_speaker_airtime_sec 9.0, stitch_gap_sec 2.0. Energy-based VAD fallback when no Hugging Face token was present.
ASR	bengaliAI/tugstugi_bengaliai-asr_whisper-medium, revision da605cc1bd2f60a18d8e440e977ddfa921a88e63, CTranslate2 runtime; beam_size 5, repetition_penalty 0.8, max_chunk_sec 28.0; per-word probabilities retained
Fusion	Schema 2.0; word $\rightarrow$ speaker assignment midpoint_then_max_overlap; overlap policy first-speaker-wins
Labelling	Claude Opus 5 under a fixed rubric; structured JSON output schema
Modelling	csebuetnlp/banglabert, revision 9ce791f3; PyTorch 2.12, transformers, scikit-learn; hand-written training loop

6.4 Variables, measurements and metadata

Per episode the registry records `episode_id`, channel, programme title, duration (s), sample rate, channels, codec, bit rate, file size, SHA-256 and a licence-attestation field. Per word the fused files record start/end time (s), the token, an ASR probability, the assigned `speaker_id` and the assignment method. Per turn the corpus records the identifiers, timing, text, word count, mean ASR confidence, diarization provider and trust flag, quality flags and the inferred role. The complete field list is in Section 7.2 and Appendix A.

6.5 Annotation / labeling protocol

Labels were assigned by a Claude Opus 5 judge, one call per speaker-episode unit, returning the speaker verdict *and* every turn label in the same call so that turn labels are consistent with the holistic verdict. All 150 eligible units were judged. The rubric encodes the failure modes of this genre:

- **Evaluative stance, not emotion.** A calm description of a banking collapse is negative.
- **Direction words are not polar.** “Increased” is positive for reserves and negative for inflation; polarity follows the orientation of the target.
- **A pointed question is neutral.** Hosts elicit, they do not assert.
- **A recommendation is neutral; a complaint is negative.**
- **Reported speech is not the speaker’s own stance.**

- `illegible` is a first-class label for ASR-destroyed turns, which are then dropped from supervision rather than guessed.

Each turn label carries a confidence and an **evidence quote**. We measured how literal these quotes are: **58% occur verbatim in the turn they cite and 42% are close paraphrases of it** (59.4% in the first judging round, 56.9% in the second, so this is a stable property of the judge rather than a regression). The quotes therefore localise a label to a specific turn and claim, which is what makes manual spot-checking fast, but they are *not* exact-match provenance and must not be used as string keys. The measurement script is released with the corpus. **The shipped labels are a single judging pass**, produced in an interactive session rather than through the batch API, so `pass_agreement` equals 1.0 by construction and no self-consistency figure exists yet. Multi-pass judging (`-passes 3`) and the 180-turn human verification sample are specified in the repository and are the designated next step (Sections 9 and 12).

Re-judging after the second pipeline run. When the bundles were re-processed, adding twenty episodes and re-running `pyannotate` over roll 2107009, the turn segmentation changed underneath most of the existing labels. A judgement was carried forward *only* if its speaker key still existed *and* its ordered list of `turn_ids` was byte-identical to the rebuilt corpus. Fifty-three of the original judgements passed that test; the remaining ninety-seven units were judged again from scratch against the new segmentation. The merge script enforces the check programmatically and asserts a zero turn-count mismatch across all 150 released blocks, so no label is silently attached to a turn it was not written for.

6.6 Human review and correction

The labels are machine-authored but the corpus is not machine-accepted. Every stage between the raw broadcast and the released supervision set passed through a human decision, and each of those decisions left an artefact in the archive that a reader can check:

- **The rubric is human-written.** The six genre-specific rules above were written by the team after reading transcripts, not derived by the model. The judge's decisions are therefore constrained by human criteria that are published in full.
- **Diarization was audited and re-run.** Inspection of build 1 showed that roll 2107009 had been segmented by the fallback provider into unusable fragments; ep001 alone produced 263 of them. The episode set for that roll was re-diarized with `pyannotate` and ep001 resolved to 47 real speaker turns. All 45 episodes in the released build carry `diarization_trusted = true`; none did in build 1.
- **Units were screened before judging.** Of 212 reconstructed speaker units, 62 were rejected as unjudgeable and never sent, for untrusted diarization, an unassigned speaker, or fewer clean words than the human-set minimum, leaving 150. A further three were withdrawn from the speaker-level task after review, giving the 147 units carrying `speaker_level_valid = true`.
- **Out-of-domain material was caught by reading, not by a filter.** Two episodes were found on inspection to be off-genre (a law-and-order interview; a news bulletin spliced onto a talk show). Both are labelled `illegible` throughout so they contribute no supervision while remaining in the corpus for transparency.
- **Stale labels were detected and discarded, not repaired.** When the corpus was rebuilt, a judgement was carried forward only if its ordered `turn_id` list was byte-identical to the new segmentation. Fifty-three passed; ninety-seven units were re-judged from scratch and nineteen stale blocks were dropped. The merge script asserts zero turn-count mismatch across all 150 released blocks.
- **ASR-destroyed turns are excluded rather than guessed.** 181 of the 1,811 turn labels are `illegible`

and are dropped from supervision, leaving 1,630 supervised turns.

- **Evidence quotes were audited.** All 715 evidence quotes were checked against the turns they cite; 58% are verbatim spans and 42% close paraphrases. The measurement script ships with the corpus so the audit is repeatable.

What this review does *not* include, stated explicitly. No independent human annotator re-labelled individual turns. The per-episode annotation drafts in the roll folders (`annotation_drafts/`, 46 files, 5,395 utterances) were prepared for exactly that step and every one of them still carries `human_verified: false`, `annotation_status: review_required`, an empty `labels` list and an empty `annotator_id`. Human effort in this build went into deciding *what was fit to label* and *what had to be thrown away*; it did not go into overruling the judge on individual turns. Consequently every accuracy figure in this report is agreement with the judge, not with human truth, and the 180-turn dual-annotated verification sample specified in Section 12 remains the first item of future work.

6.7 Provenance, versioning and raw-data preservation

Raw audio is preserved unmodified in the roll folders and is never overwritten; all derived artefacts are written to a separate `results/` tree per roll. A `run_manifest.json` records the full configuration and a SHA-256 for every produced file, and a `models/manifest.json` records the model URI, commit hash and notes for each network used. Identifiers are hierarchical and stable: `roll::episode::turnNNNN`, with `global_episode = roll::episode` used as the grouping key for splitting.

7. Data Description / Data Records

Explain exactly what a user will find in the released dataset. The folder/file structure should be understandable without opening every file. Include raw and processed components separately.

7.1 A worked example: one moment of audio, end to end

Before the file listings, Figure 3 follows a single moment of one episode through every stage of the release, so that a reader can see what the data actually contains rather than only how it is filed. Nothing in the figure is retyped: `make_sample_figure.py` reads it straight out of the released files, so it cannot drift from the corpus.

Three properties of the corpus are visible in that single figure. A decode-window boundary is not a speaker boundary, which is why the windows are merged back into turns before anything is labelled. Turn labels carry a cited span, so a reader can check a label against the text without listening to the audio. And the speaker verdict is a separate judgement, not a tally of the turn labels beneath it.

7.2 Repository / Folder Structure

Folder / file	Contents	Raw / processed / metadata
<code><roll>/*.mp3, *.mpeg</code>	Source broadcast audio as collected (not redistributed)	Raw
<code><roll>/results/ raw/</code>	<code>registry.csv</code> , <code>episode_metadata.csv</code> : per-episode provenance and checksums	Metadata

Folder / file	Contents	Raw / processed / metadata
<roll>/results/processed/	16 kHz mono WAV per episode; hygiene_report.json	Processed
<roll>/results/diarization/	epNNN.json, epNNN.rttm, epNNN_raw.rttm: speaker segments	Processed
<roll>/results/asr/	epNNN.json: transcripts with per-word timings and probabilities	Processed
<roll>/results/fused/	epNNN.json: word → speaker aligned utterances (schema 2.0)	Processed
<roll>/results/drafts/, annotated/	Annotation drafts and review status	Metadata
<roll>/results/aggregated/	submission_gates.json automated quality gates	Metadata
<roll>/results/artifacts/, models/	run_manifest.json, manifest.json: config, checksums, model provenance	Metadata
speaker_sentiment/data/	turns.jsonl, speakers.jsonl, judgements.jsonl, turns_labelled.jsonl, speakers_labelled.jsonl, turns_{train,val,test}.jsonl, speakers_{train,val,test}.jsonl	Processed / ML-ready
speaker_sentiment/*.py	build_speaker_corpus.py, llm_judge.py, make_datasets.py, predict_speakers.py	Code
speaker_sentiment/*.ipynb	Speaker_Sentiment_BanglaBERT.ipynb: training, aggregation, ablation, verification	Code
speaker_sentiment/runs/	Trained model, metrics.json, test_predictions.jsonl	Results
sentiment/lexicon/	financial_targets.csv (52 targets), polarity_cues.csv (103 cues)	Metadata
README.md	Access, pipeline, caveats and run instructions	Documentation

7.3 Data Dictionary

Representative fields of `turns.jsonl`, the primary release file. The complete dictionary for all release files is in Appendix A.

Variable	Meaning	Type	Unit / range	Allowed values	Missing rule
turn_id	Unique turn identifier	string	n/a	roll::ep::turnNNNN	never null
global_episode	Episode grouping key used for splitting	string	n/a	roll::epNNN	never null
speaker_id	Diarization cluster index within the episode	integer	≥ -1	-1 = unassigned	-1
start_sec, end_sec	Turn boundaries in the episode	float	seconds	≥ 0	never null
text	Normalised Bangla transcript of the turn	string	n/a	free text	" " if empty
n_words	Whitespace word count of text	integer	≥ 0	n/a	never null

Variable	Meaning	Type	Unit / range	Allowed values	Missing rule
asr_ confidence	Mean log-probability over the turn's words	float	≤ 0	n/a	null if no words
diarization_ trusted	False for the five fallback-diarized episodes	boolean	n/a	true/false	never null
role	Inferred conversational role	string	n/a	host, guest, minor, unassigned	unassigned
label	Judge turn label (labelled files only)	string	n/a	negative, neutral, positive	row dropped if illegible
judge_ confidence	Judge's stated confidence in the turn label	float	[0, 1]	n/a	never null
evidence	Verbatim quote supporting the label	string	n/a	substring of text	" "

If the dataset has many variables, keep only representative rows here and place the complete data dictionary in Appendix A and/or the repository.

7.4 Dataset Composition and Descriptive Overview

Provide concise descriptive information needed to understand dataset composition: counts by class/category, number of subjects/sources, temporal/geographic coverage, missingness summary, and representative examples. Do not turn this into a full research-results section.

Property	Value
Unique episodes (after transcript-hash dedup)	45 (from 46 processed, 56 registry rows)
Episodes with trusted (pyannote) diarization	45, all of them
Episodes with fallback diarization	0 (was 5 in build 1; re-diarized)
Total audio	≈ 28.6 h
ASR decode windows	5,327
Speaker turns	2,414 ($\approx 219,000$ words)
Turns by role	guest 1,236, host 926, unassigned 233, minor 19
Speaker-episode units	212 total; 150 judged; 147 speaker-level valid (45 hosts, 103 guests, 2 minor)
Turn labels produced	1,811; illegible 181 (10.0%) dropped
Supervised turns	1,630 : negative 53.4%, neutral 34.7%, positive 11.9%
Speaker verdicts	negative 110, neutral 18, positive 11, mixed 8
Human sentiment labels	0 (see Sections 9 and 12)

Role structure separates cleanly and confirms the rubric behaved as designed: hosts take many short turns (mean 39 words) and are majority `neutral`, while guests take few long turns (mean 146 words) and carry the evaluative content. Table 1 gives the breakdown.

8. Data Preprocessing, Curation and Leakage Control

Document every transformation from raw data to the released/ML-ready data. Preserve the distinction between data cleaning and model-specific preprocessing.

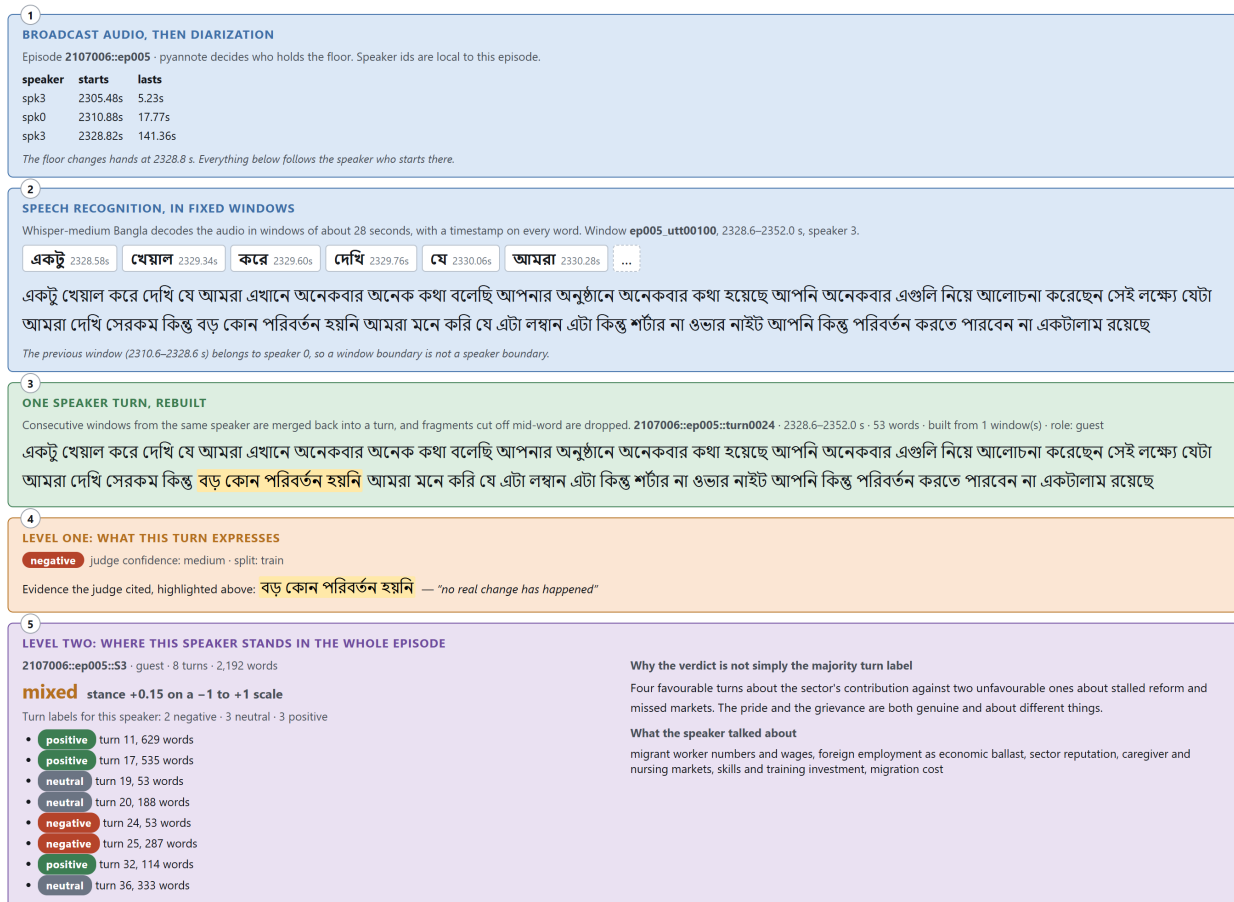


Figure 3: One moment of 2107006 : : ep005, from audio to speaker verdict. Diarization hands the floor to speaker 3 at 2328.8s; Whisper decodes the next window with a timestamp on every word; the window is merged back into a speaker turn; the judge labels that turn *negative* and cites the span highlighted in the text; and the speaker's verdict for the whole episode is *mixed*. The last two disagree on purpose. This guest is favourable about what migrant labour contributes and unfavourable about stalled reform, and the release records both levels rather than forcing one of them.

Table 1: Turn-label distribution by inferred speaker role over the 1,630 supervised turns. Host neutrality is the expected consequence of the rubric rule that a pointed question is elicitation rather than assertion. The pattern held after the corpus grew by 22%, which is evidence the rubric was applied consistently across both judging rounds.

Role	Negative	Neutral	Positive	Total
Guest	602	192	171	965
Host	266	374	23	663
Minor	2	0	0	2
Total	870	566	194	1,630

- **Cleaning.** One duplicate episode was removed by transcript hash. Repetition-loop turns, an ASR failure mode in which one syllable repeats for hundreds of characters, are detected and flagged `flag_repetition_loop`. Turns below a minimum length are flagged `flag_too_short`.
- **Window → turn reconstruction.** The “utterances” in the fused files are *not* turns: they are fixed ≈ 28 s ASR decode windows, a large fraction of the 5,327 having a duration of 27.2–28.0 s. Every window boundary truncates a word, so many windows begin with an orthographically impossible Bengali dependent sign. Consecutive same-speaker windows are therefore merged back into turns and orphaned fragments dropped.
- **Missing data.** Words that fusion could not attribute carry `speaker_id = -1`; the corresponding turns are retained with `role = unassigned` and a `diarization_trusted` flag, rather than deleted or imputed. Missingness is flagged, never silently filled.
- **Outliers / noise.** Turns the judge marked illegible (181, 10.0%) are retained in `turns.jsonl` but excluded from `turns_labelled.jsonl` and all splits, so the raw record is preserved while supervision stays clean.
- **Encoding / normalisation (model-specific).** Unicode NFC normalisation, with the `csebuetnlp` normalizer applied where available because BanglaBERT was pretrained on normalised text. Inputs are constructed as `role_turn` sentence pairs and tokenised at `max_len` 128 on CPU or 256 on GPU; 256 truncates $\approx 10\%$ of turns and 128 truncates $\approx 23\%$ (subword length `p50 = 46`, `p75 = 120`, `p90 = 254`, `p95 = 351`, `max = 1,265`).
- **Augmentation / synthetic data.** None. No synthetic turns or oversampled copies were created; class imbalance is handled by class weighting in the loss, not by resampling.
- **De-identification.** Speakers are released as episode-local cluster indices (`Speaker 0`, `Speaker 1`, ...) and are never linked to named individuals. Speaker IDs are model predictions, not person identities.
- **Leakage control.** The grouping key is `global_episode`. Splits are constructed episode-disjointly and `make_datasets.py` asserts disjointness rather than assuming it, because two turns from one talk show share the panel, the topic, the presenter’s phrasing and the same ASR error signature, so a random row split would let the model recognise the episode instead of the stance. The duplicate episode was removed *before* splitting, so the same panel cannot appear on both sides. **Known residual risk:** a programme-level split is the stricter control, but a majority of episodes carry a placeholder or empty programme title in the registry, so grouping by programme collapses the corpus; the split therefore warns that train and test share the programme label `business talk`, and this is accepted knowingly (Section 12).

9. Technical Validation and Data Quality

This section supports the technical quality of the dataset. Validation should concern the data, not merely model accuracy.

Quality dimension	Evidence / checks performed
Completeness	56 registry rows → 46 processed episodes → 45 unique. Every retained episode has diarization, ASR, fusion and draft artefacts; per-episode counts are reconciled in the <code>stats</code> block of each fused file. Automated gates in <code>submission_gates.json</code> record which checks pass, fail or are skipped.
Consistency	Fused files validate against schema 2.0. Turn counts, word counts and durations are recomputed from the fused files at corpus-build time and cross-checked. Identifier format is enforced; episode disjointness across splits is asserted programmatically.

Quality dimension	Evidence / checks performed
Label quality	Single-pass LLM judge with a fixed, byte-identical rubric and structured JSON output. Every turn label carries a confidence and an evidence quote; 58% of the 715 quotes are verbatim spans of the cited turn and 42% are close paraphrases (measured, both rounds alike), so labels are spot-checkable but the quotes are not exact provenance. Internal coherence measured: rebuilding each speaker verdict bottom-up from its own turn labels, without showing the aggregator the holistic verdict, reproduces it 85.0% of the time exactly (<code>word_weighted</code>) and 87.1% by polar direction (<code>majority_vote</code>). Coherence <i>rose</i> from 78.7%/83.6% when the corpus grew from 72 to 150 judged units, which is evidence of a stable rubric rather than drift. Not yet measured: inter-pass self-consistency (labels are one pass) and agreement with human annotators.
Measurement quality	Fixed decoding settings (beam 5, repetition penalty 0.8, 28 s chunks) and fixed diarization thresholds across all episodes. Per-word ASR probabilities and per-turn mean confidence are retained so low-confidence material can be filtered downstream. Audio hygiene reports are stored per episode.
Integrity	SHA-256 recorded for every source file in <code>registry.csv</code> and for every produced file in <code>run_manifest.json</code> ; <code>sha256_verified</code> is true for all registry rows. Exact-duplicate detection is by transcript hash.
Representativeness / bias	Class imbalance persists (positive is 11.9% of supervised turns, up from 10.2%). Strong negative prior (110 of 150 speaker verdicts). This is <i>partly real</i> , since the episodes cover a banking crisis, a reserve crisis, an energy shock and inflation, and <i>partly selection and possible judge drift</i> ; the two explanations are separated only by human verification. Diarization is now uniform: all 45 episodes are <code>pyannote</code> . Two episodes contain out-of-domain material (one law-and-order interview; one news bulletin spliced onto a talk show), and both are labelled <code>illegible</code> throughout so they never enter supervision.
Reproducibility	<code>build_speaker_corpus.py</code> regenerates <code>turns.jsonl</code> and <code>speakers.jsonl</code> from the fused bundles in ≈ 20 s; <code>make_datasets.py</code> regenerates all six split files in ≈ 5 s with an explicit seed (13). Pipeline seed 17 and every model revision hash are recorded in <code>run_manifest.json</code> . The baseline trains on CPU in ≈ 60 min, so results are re-derivable without a GPU.

Gates that remain open, reported honestly. `submission_gates.json` records `wer_measured`, `der_measured` and `annotation_kappa` as SKIPPED because no gold transcript, gold RTTM or second annotation file exists. Every number in Section 10 is therefore bounded by an unmeasured word error rate.

9.1 Comparison with Existing Datasets and Significance

Compare the dataset with similar, related and competitive datasets with proper citation in tabular form.

Four properties decide whether a resource can answer the question in Figure 1(a). It has to be **Bangla**; it has to be **spoken** material rather than writing; it has to keep **who said what** instead of pooling a conversation into one text; and it has to carry a **verdict on the person**, not only on each sentence. The table below scores the nearest resources on those four. Each is strong on some and silent on others; the gap this dataset fills is the row where all four are present at once.

Dataset	Bangla	Spoken	Speaker kept apart	Verdict on the person	What it gives, and what it cannot
BLUB / BanglaBERT [1]	Yes	No	No	No	Supplies the encoder this work fine-tunes. Written text, one label per document.

Dataset	Bangla	Spoken	Speaker	Person verdict	Relation
BLP-2023 Task 2 [2]	Yes	No	No	No	The closest Bangla sentiment benchmark. Social-media posts have a single author and no floor to share.
BAN-ABSA [3] / BABSA [4]	Yes	No	No	No	Ties sentiment to a target, which this work also needs, but the target is an aspect in written text, never a participant in a conversation.
MELD / IEMOCAP [5]	No	Yes	Yes	No	Proves conversational, speaker-aware labelling works. English, scripted or acted, and the label stops at the utterance.
Earnings-call corpora [6]	No	Yes	Yes	Partly	Nearest in intent: financial speech attributed to named executives. English, professionally transcribed, so it never tests robustness to ASR error.
This dataset	Yes	Yes	Yes	Yes	Bangla broadcast debate, ASR-noisy, 2,414 turns kept per speaker, 147 speaker-episode verdicts with a stance score.

The distinction in one sentence. Every Bangla sentiment resource we could find labels writing, and every speaker-aware conversational resource we could find is in English and stops at the utterance; this dataset is the intersection, and it adds the level neither side has, which is a verdict on a participant across a whole episode.

Identify and mention in points the novelty and significant properties of the dataset w.r.t. existing datasets.

- **Two supervised tasks, not one.** The release supports turn sentiment (1,630 labelled turns) and a separate verdict on each speaker across a whole episode (147 units with a stance score). Both are given a full evaluation in Section 10, with baselines, confusion matrices and slice breakdowns for each.
- **The second task is measurably not reducible to the first.** Both levels come from one judging call over the speaker’s full script, so they are coherent by construction, and rebuilding the verdict from its own turn labels reproduces it 85.0% of the time. The remaining 15% is the point: an oracle holding the gold turn labels still cannot recover those verdicts, so roughly one speaker in seven can only be judged from the episode. That is the headroom a speaker-level model exists to capture, and no utterance-level corpus exposes it.
- **Speaker-episode as a released unit.** Existing conversational sentiment corpora stop at the utterance; this release ships the speaker unit with a continuous stance score, which is the unit financial analysis actually needs.
- **First Bangla speaker-attributed sentiment resource in broadcast financial speech,** to the best of our search; the existing Bangla resources are written text.
- **Genre-specific rubric.** The host-question-is-neutral and target-orientation rules are specific to panel discussion and to economic vocabulary, and the role split in the released labels demonstrates they took effect.
- **Full intermediate provenance.** RTTM, per-word ASR probabilities, fusion alignments and judge rationales are all retained, so the corpus can be re-derived under a different diarizer, ASR model or labelling rule.

- **Documented defects, and a documented repair.** The duplicate episode, the out-of-domain material and the placeholder programme titles are published rather than concealed. Two defects reported in build 1 have since been fixed: roll 2107009 was re-diarized with `pyannote`, and the degenerate speaker-level validation split (six units over two classes) now holds 23 units over all four classes.

10. Machine Learning (ML) Use Demonstration

The purpose is to demonstrate that the dataset can support a valid ML workflow. Keep the emphasis on dataset usability and reproducibility rather than claiming a new algorithm.

10.1 ML task definition

The dataset defines **two tasks, not one**, and they are evaluated separately because they are different problems.

Task A, turn sentiment. Input X is the pair (role, turn text); the target $y \in \{\text{negative, neutral, positive}\}$; the unit of prediction is one speaker turn. There are 1,630 supervised turns.

Task B, the speaker’s position across the whole episode. Input is everything one speaker said in one episode; the target $y \in \{\text{negative, mixed, neutral, positive}\}$ together with a continuous stance score in $[-1, 1]$; the unit of prediction is one speaker–episode unit. There are 147 such units.

Why Task B is not Task A with a summing step. A verdict on a person is not the average of their sentences. A panellist can concede a point warmly and still hold a negative position; a host can sound severe while taking no position at all; and the `mixed` verdict has no turn-level counterpart, because it describes a speaker who is genuinely favourable about one thing and unfavourable about another. Figure 3 shows a real case from the release: a guest whose eight turns are three positive, three neutral and two negative, and whose episode verdict is `mixed` with a stance of +0.15. The Task B results below quantify the gap. Even an oracle that is handed the judge’s own gold turn labels and combines them perfectly recovers only 85.0% of the speaker verdicts, so roughly one verdict in seven is not present in the turn labels at all and has to be read from the episode. Task A therefore cannot subsume Task B, and the report measures both.

10.2 Data partitioning strategy

Episode-grouped split, **never** a row split, with seed 13 and the configuration `-group-by episode -test-episodes 9 -val-episodes 7`. Seven candidate seeds were compared on class balance before one was fixed; seed 13 is the only one placing all four speaker classes and a workable share of positive turns in every partition. Table 2 gives the partition.

Table 2: Episode-disjoint partition of the supervised corpus (seed 13). All three splits carry all three turn classes and all four speaker classes; the degenerate validation split reported in build 1 no longer occurs.

Split	Episodes	Speakers	Turns	Neg.	Neu.	Pos.
Train	29	94	1,009	530	365	114
Val	7	23	297	162	93	42
Test	9	30	324	178	108	38

Leakage rationale: turns from one episode share the panel, topic, presenter phrasing and ASR error signature. Disjointness is asserted in code. The duplicated episode was removed before splitting. Speaker-level, the partition holds 94 / 23 / 30 valid units; the validation split now contains negative 16, neutral 3, positive 2 and mixed 2, so speaker-level validation is meaningful for the first time.

10.3 Preprocessing and feature/representation steps

Unicode NFC plus the `csebuetnlp` normalizer; `role_turn` sentence-pair construction; BanglaBERT WordPiece tokenisation at `max_len` 128 (CPU run) or 256 (GPU). No vocabulary, scaler or statistic is fitted on validation or test data: the tokenizer is pretrained and the class weights are computed on the training split only.

10.4 ML model(s) and hyperparameters

Majority-class baseline	Predicts the most frequent training class; reported as the floor.
TF-IDF + logistic regression	Word 1–2-grams (<code>min_df=2</code> , sublinear TF), balanced class weights. The feature family (word 1–2, <code>char_wb</code> 2–5, <code>char_wb</code> 3–5) and $C \in \{0.5, 1, 2, 4, 8\}$ were selected together on the validation split; word 1–2 with $C = 2$ won at 0.578. Reproducible with <code>tfidf_baseline.py</code> .
BanglaBERT (main)	<code>csebuetnlp/banglabert</code> (ELECTRA discriminator, 110M), <code>input_mode = role_turn</code> , <code>max_len</code> 256, 6 epochs (best epoch 4, restored by validation macro-F1), batch 16, AdamW with encoder LR 2×10^{-5} and head LR 1×10^{-4} , linear warmup 10%, weight decay 0.01, gradient clipping 1.0, label smoothing 0.1, class weighting on (positive is 11.9% of the data), per-sample weights from judge confidence $\times$ pass agreement, seed 42. Trained on a Colab T4. Hand-written training loop rather than <code>Trainer</code> , so the two weighting schemes compose cleanly.

10.5 Evaluation metrics

Macro-F1 is the headline metric because the corpus is strongly imbalanced and the positive class, the one a financial sentiment index most needs, is the rarest. Reported alongside: per-class precision/recall, the majority-class macro-F1 floor, and, at the speaker level, exact verdict accuracy, polar-direction accuracy, and Pearson correlation between the predicted and judged stance scores.

10.6 Reproducibility

Notebook `speaker_sentiment/Speaker_Sentiment_BanglaBERT.ipynb`; corpus rebuilt by `build_speaker_corpus.py` and `make_datasets.py`; CLI scoring by `predict_speakers.py`, which reads `input_mode`, `max_len` and the winning aggregation rule out of the written `metrics.json` so the CLI cannot silently disagree with the model that produced it. Split seed 13, training seed 42. The reported run was executed on a Colab T4 GPU; `history.csv`, `metrics.json`, `test_predictions.jsonl` and the `presentation/` artefacts in `runs/banglabert_turn/` are released so every figure and table below can be recomputed without retraining. Measured runtime: ≈ 30 min for 6 epochs at `max_len` 128 on CPU (10 threads), ≈ 60 min at 256; ≈ 3 min on a Colab T4. On Windows, `python -X utf8` is required.

10.7 Results (compact)

Table 3: Turn-level results on the episode-disjoint test split (324 turns). Macro-F1 is the headline metric because the positive class is only 11.9% of the supervised corpus. Both learned models were selected on the validation split and evaluated once on test.

Model	Val macro-F1	Test macro-F1	Test acc.	κ	Notes
Majority class	n/a	0.236	0.549	0.000	Floor
TF-IDF word 1–2 + LogReg	0.578	0.529	0.623	0.339	Non-neural reference; $C = 2$
BanglaBERT (<code>role_turn</code>)	0.704	0.670	0.701	0.496	Main baseline; class-weighted

Table 4: Per-class turn-level performance of the BanglaBERT baseline on the test split. The rarest class, `positive`, is the weakest but remains far above chance; the class-weighted loss is what keeps its recall usable.

Class	Precision	Recall	F1	Support
negative	0.790	0.697	0.740	178
neutral	0.655	0.722	0.687	108
positive	0.521	0.658	0.581	38
Macro avg	0.655	0.692	0.670	324
Weighted avg	0.713	0.701	0.704	324

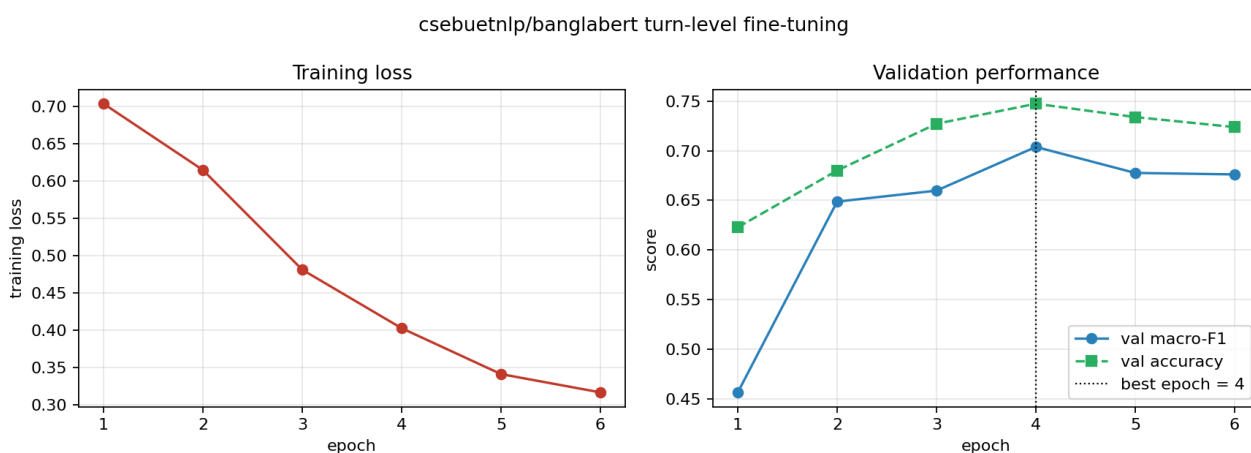


Figure 4: Training loss and validation performance per epoch. Validation macro-F1 peaks at epoch 4 (0.704) and then declines while training loss keeps falling. The model begins to overfit the 1,009 training turns, and the epoch-4 checkpoint is the one restored and evaluated.

Where the errors are. Of 97 test errors, 37 are `negative` turns predicted `neutral` and 24 are `neutral` predicted `negative`; the `positive/negative` confusion is far rarer (17 and 9 turns). The model therefore recovers *polarity direction* more reliably than it recovers *intensity*, which matters for the intended application: a sentiment index built on this output would rank speakers correctly more often than it would place them on an absolute scale.

10.8 Task B results: the speaker’s position across the whole episode

This is the unit the dataset was built for, and it is reported with the same apparatus as Task A: a baseline ladder, a confusion matrix, per-class scores, slice breakdowns and an error analysis.

Protocol. All 147 valid speaker units are scored under 5-fold `GroupKFold` grouped by episode, so no episode contributes to both sides of a fold and every unit is held out exactly once. Every system in Table 6 sees the identical fold assignment. This replaces the 30-unit held-out split used for Task A, which is too small to separate four classes: a single unit there moves accuracy by more than three points.

Read macro-F1, not accuracy. 110 of the 147 units are `negative`, so a model that answers `negative` every time is already 74.8% accurate while being useless. Table 6 shows exactly this trap. The TF-IDF control posts the *highest* accuracy of any learned system (0.775) and the *second-lowest* macro-F1 (0.342), because it mostly reproduces the prior. The hierarchical model gives up 12 points of accuracy and gains 19 points of macro-F1, which is the trade a speaker-resolved index actually wants: it is the only system that identifies the minority positions an analyst is looking for.

The second route: aggregating Task A predictions. A speaker verdict can also be reached without a speaker model at all, by applying a fixed rule to the turn labels the Task A classifier predicts. This is the route the

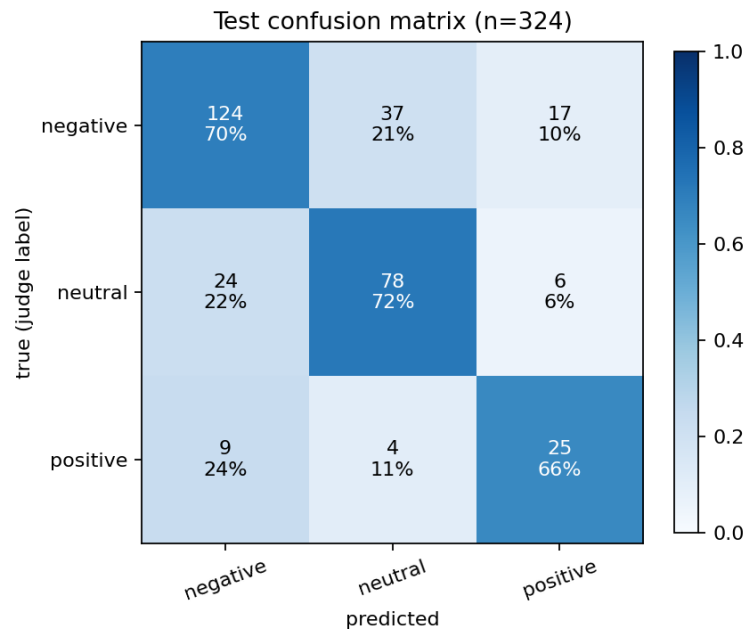


Figure 5: Test confusion matrix (counts and row-normalised). The dominant error is `negative` misread as `neutral` (37 turns): understated criticism in a panel register, which is exactly the distinction the rubric was written to enforce.

Table 5: Test accuracy by speaker role and by ASR-confidence quartile. Host turns are easier to classify but score lower on macro-F1 because they are overwhelmingly `neutral`, so the two rare classes carry the macro average. Performance degrades gracefully as transcript quality falls: the worst ASR quartile loses about 8 points of macro-F1 relative to the best.

Slice	n	Mean ASR conf.	Accuracy	Macro-F1
<i>by inferred role</i>				
Guest	184	n/a	0.696	0.610
Host	140	n/a	0.707	0.468
<i>by ASR confidence quartile (mean log-probability per word)</i>				
Q1 (worst)	83	−0.061	0.675	0.651
Q2	79	−0.044	0.709	0.658
Q3	82	−0.037	0.683	0.622
Q4 (best)	80	−0.028	0.738	0.729

Table 6: Task B systems over all 147 speaker units, 5-fold episode-disjoint cross-validation. The oracle is not a competitor: it is handed the judge’s own gold turn labels and combines them word-weighted, so it bounds what any turn-aggregation route can reach. Its 85.0% ceiling is the measured reason Task B needs its own model.

System	Accuracy	Macro-F1	Notes
Majority class	0.748	0.214	always predicts <code>negative</code>
TF-IDF over the speaker’s script	0.775	0.342	bag-of-words control
Hierarchical BanglaBERT	0.653	0.530	attention pooling over turn embeddings
<i>Oracle: gold turn labels aggregated</i>	<i>0.850</i>	<i>0.664</i>	ceiling, not a model

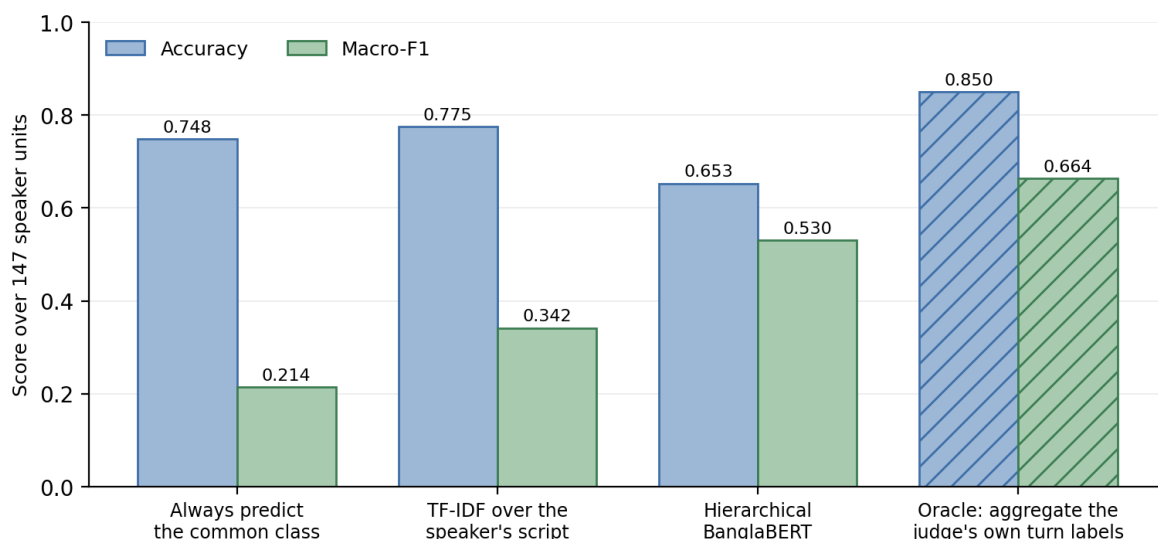


Figure 6: Task B, the same four systems. Accuracy and macro-F1 move in opposite directions across the ladder, which is why accuracy alone would rank the weakest system first. The hatched pair is the oracle ceiling.

shipped command-line scorer uses, and it is reported on the 30-unit held-out split so that it lines up with the Task A evaluation. Table 7 gives it. The figures there look higher than Table 6 and are not comparable with it: 30 units against 147, a single split against cross-validation, and a rule chosen on train+val rather than learned. It is included because it is what the tool does, not as a rival measurement.

Table 7: Speaker-level results on the 30 valid test speaker units, by aggregation rule. The rule was selected on the 117 train+val units, where `hard_vote` won (exact accuracy 0.752), and that rule is the one reported on test. `top_k` transfers to a higher macro-F1, but quoting it would be selection on the test set.

Aggregation rule	Exact acc.	Direction acc.	Stance corr.	Macro-F1
hard_vote (selected)	0.867	0.867	0.451	0.567
<code>top_k</code>	0.800	0.800	0.628	0.667
<code>word_weighted</code>	0.767	0.800	0.569	0.480
<code>dur_weighted</code>	0.767	0.800	0.567	0.480
<code>unweighted</code>	0.633	0.667	0.691	0.447

Table 8: Hierarchical speaker model (frozen turn encoder + attention pooling), out-of-fold over **all 147 valid speaker units** under 5-fold `GroupKFold` grouped by episode. Because every labelled speaker is used as test data exactly once, this is the speaker-level figure the report quotes; the 30-unit held-out split in Table 7 is a sanity check, not a measurement.

Class	Precision	Recall	F1	Units
negative	0.882	0.682	0.769	110
neutral	0.519	0.778	0.622	18
positive	1.000	0.455	0.625	11
mixed	0.067	0.250	0.105	8
Overall: accuracy 0.653, macro-F1 0.530 (folds 0.500–0.828)				

Where Task B fails. Of 51 errors, exactly half (25) are `negative` speakers called `mixed`. The `mixed` class is the problem: 8 training examples across the whole corpus, an F1 of 0.105, and 30 predictions made where only 8 were warranted. It is over-predicted because a speaker with any favourable turn at all looks `mixed` to the pooling layer. The other direction is clean: no speaker is ever called `positive` wrongly, and `neutral` recall is 0.778. Collapsing `mixed` into the polar classes would raise the headline, so it is worth saying plainly that we did not do that. The class is small because genuinely two-sided speakers are rare, not because the label

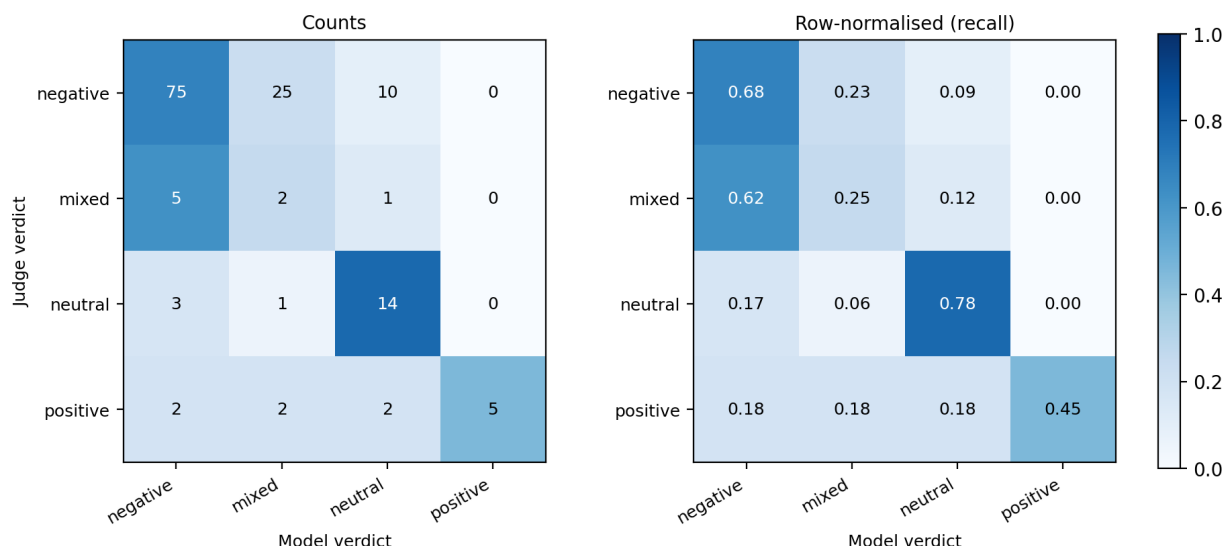


Figure 7: Task B confusion matrix over the 147 out-of-fold speaker verdicts. The model never labels a speaker `positive` unless it is right (precision 1.000), but it reaches only 45% of the positive speakers. The costly error is the opposite one: 25 genuinely negative speakers are called `mixed`, which is the model hedging on a class it has only eight examples of.

is ill-defined, and Figure 3 shows one being applied correctly.

Table 9: Task B accuracy and macro-F1 by slice. The model is weakest on the speakers who talk most, which is the opposite of the Task A pattern and is consistent with the `mixed` confusion above: the longer a speaker holds the floor, the more likely they say something favourable somewhere, and the more likely the pooling layer hedges.

Slice	Units	Accuracy	Macro-F1
<i>by inferred role</i>			
Host	45	0.689	0.463
Guest	101	0.644	0.381
<i>by how much the speaker said</i>			
1–4 turns	28	0.643	0.531
5–12 turns	74	0.662	0.607
13+ turns	45	0.644	0.324
<i>by transcript quality</i>			
good and fair	144	0.660	0.542
poor	3	0.333	0.167

What the two tasks together say about the corpus. Task A recovers turn sentiment at 0.670 macro-F1 against a 0.236 floor, so the turn labels are learnable. Task B recovers the speaker verdict at 0.530 against a 0.214 floor, so the speaker labels are learnable too, and by a wider relative margin, but it sits well under the 0.664 oracle. The distance between 0.530 and 0.664 is what a better Task B model could still win by reading the turns more carefully; the distance between 0.664 and 1.000 is what no turn-aggregation method can win at all, and is the part of the speaker verdict that lives in the episode rather than in any single turn. Both figures are agreement with the judge, not with human ground truth.

Reproducing Task B. `speaker_level_eval.py` recomputes every number in this subsection, including the baselines, from the released out-of-fold predictions in about 15 seconds on CPU; `make_speaker_figs.py` redraws the two figures.

The shipped scorer reproduces these numbers. The command-line tool `predict_speakers.py` reads the rule, input mode and `max_len` from `metrics.json` so it cannot silently disagree with the notebook that

trained the model. Run end-to-end on CPU over the 324 held-out turns it recovers all 30 test speaker units and reproduces the notebook’s held-out figures exactly (exact accuracy 0.867 and stance correlation 0.451 under the selected `hard_vote` rule) in 47 seconds without a GPU. All 324 turn-level predictions are identical to the ones the notebook wrote, which also shows that the optional `csebuetnlp` normaliser is not load-bearing on this corpus: the CPU run used the NFC fallback and still matched every label.

Briefly state what the baseline demonstrates about dataset usability. Avoid over-interpreting performance.

The baselines demonstrate that the corpus supports a valid supervised workflow at both levels. At the turn level a standard Bangla encoder reaches macro-F1 0.670 against a 0.236 floor and a 0.529 non-neural reference, from 1,009 training turns under an episode-disjoint split, with $\kappa = 0.496$, which is moderate agreement with the judge on a three-class problem. At the speaker level, aggregating those predictions reproduces the judge’s verdict on 26 of 30 held-out speakers, and the hierarchical model over all 147 units reaches 0.653 accuracy.

Four cautions belong with these numbers. First, **every score measures agreement with the LLM judge, not with human ground truth**; macro-F1 0.670 means the student model reproduces the judge, whose own accuracy is unmeasured. Second, `mixed` is not learnable at this scale: eight units in the entire corpus give F1 0.105, and it should be treated as unsupported until the corpus grows. Third, the speaker-level test split holds 30 units, so one unit moves macro-F1 by about 0.03 and the gap between the selected and best-transferring aggregation rule is selection noise, not a finding. Fourth, **the backbone ablation was not run**: BanglaBERT was adopted on published benchmark evidence rather than compared on this corpus, and whether BanglishBERT handles the heavily code-mixed financial vocabulary better remains an open question. The sweep is implemented in Section 10 of the notebook behind a `RUN_ABLATION` flag and costs about 25 GPU-minutes.

11. Usage Notes and Reuse Potential

Give practical instructions that help another student/researcher use the dataset correctly.

- **Loading.** Download the repository and read the JSONL files line by line (`json.loads` per line, UTF-8). For a model-ready corpus, use `turns_{train,val,test}.jsonl` directly; no rebuild is needed. To re-derive from the fused bundles, run `build_speaker_corpus.py` (≈ 20 s) then `make_datasets.py` (≈ 5 s).
- **Environment.** Python 3.10+, torch 2.12, transformers, scikit-learn, pandas, matplotlib. Install the `csebuetnlp` normalizer, because BanglaBERT was pretrained on normalised text and the fallback to bare NFC costs measurable F1. On Windows, prefix commands with `python -X utf8` or the console encoder fails on Bangla.
- **Conventions.** `speaker_id` is **episode-local**: Speaker 0 in two episodes is not the same person. `asr_confidence` is a mean log-probability, so it is negative and larger (closer to zero) is better. `stance_score` runs from -1 (most negative) to $+1$ (most positive). Group on `global_episode`, not on `roll` and not on `programme`.
- **Precautions.** Filter to `diarization_trusted = true` for any claim about speaker attribution. Filter to `speaker_level_valid = true` for any speaker-level evaluation. Exclude `2107015::ep002` from programme-level claims (out of domain). Do not tune on the speaker-level validation split. It holds 23 units, and two of its four classes have only two members each, so use `GroupKFold` over all 147 valid units instead.
- **Reuse.** The retained audio supports acoustic and multimodal extensions (`wav2vec2` / `WavLM` branches); the 52-target lexicon supports speaker $\times$ target aspect-based sentiment; the per-word ASR probabilities sup-

port ASR-robustness work on confusion networks; and the judged corpus supports LLM-judge calibration studies.

- **Inappropriate uses.** Do not present model output as the identified opinion of a named individual, because speaker IDs are clustering predictions, not identities. Do not quote the macro-F1 figures as human-validated accuracy. Do not use this corpus to make trading or policy decisions without the human-verification step in Section 12.

12. Limitations and Known Biases

Recommended: up to about 200 words. Discuss limitations of the dataset itself: sample size, coverage, imbalance, acquisition constraints, label uncertainty, measurement error, missingness, bias, or restricted generalizability. Do not hide limitations.

No human labels exist. All supervision comes from a single LLM judging pass, so `pass_agreement` is 1.0 by construction and no self-consistency or inter-annotator figure can be quoted. The judge and any model trained on it share a blind spot: if the judge over-reads negativity in garbled text, the student learns to do the same, and no evaluation against judge labels will reveal it.

Three core quality metrics are unmeasured. Word error rate, diarization error rate and annotation κ are all recorded as `SKIPPED` for want of gold references. The literature places the comfortable ceiling for sentiment on ASR output near 12% WER; this ASR is visibly worse, with invented words, dropped inflections and wrong numbers.

Scale and coverage. One hundred and forty-seven valid speaker units and 1,630 supervised turns remain small for a four-class speaker task; one unlucky episode still moves a held-out score by about three points, so grouped cross-validation and seed averaging are required before quoting a figure as final. `positive` and `mixed` number only 11 and 8 speaker units in the entire corpus. A majority of episodes carry placeholder or empty programme titles, which prevents the stricter programme-level split. Two episodes contain out-of-domain material and are excluded by labelling rather than deletion. The corpus is negative-dominant (53%), partly because the period genuinely was, with a banking crisis, an energy shock and sustained inflation, and partly through selection; these explanations are separable only by the human-verification sample.

Two build-1 limitations have been removed and are recorded here for transparency rather than deleted: `roll 2107009`'s fallback diarization (five episodes, 60–70% of words unattributed) has been replaced by `pyan-note`, and the speaker-level validation split that previously held six units across two classes now holds 23 across four.

13. Ethics, Privacy and Responsible Data Use

Complete the applicable items. If human participants or personal data are involved, state consent/approval/anonymization details. If data come from social media/web platforms, address terms of service and redistribution. If none apply, state that explicitly.

Human participants / consent	Not applicable. No human participants were recruited and no experiment was conducted. All speakers are public figures appearing voluntarily on broadcast television in a professional capacity. No ethics-board approval was sought or required for analysis of publicly broadcast material.
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Privacy / de-identification	Speakers are released only as episode-local cluster indices (<code>Speaker 0</code> , <code>Speaker 1</code> , ...) and are not linked to names. No personal identifiers, contact details or non-public information are stored. Speaker identity is a model prediction, not an assertion about a real person, and downstream users are instructed not to present it as one.
Web / social-media data	Not applicable in the social-media sense; the material is television broadcast audio obtained from public platform postings. Episodes were downloaded from publicly posted broadcast uploads with a standard command-line downloader and converted to 16 kHz mono WAV locally. Compliance with each platform's terms of service has not been formally verified , which is precisely why the audio is not redistributed and every <code>license_attestation</code> field still reads UNVERIFIED.
Copyright / third-party data	Source audio is third-party copyrighted broadcast content. The audio is not redistributed. The release contains derived artefacts (transcripts, timings, speaker segments and annotations) together with the episode registry (titles, channels, durations, SHA-256) so that a user can obtain the source independently. The derived annotations are released under CC BY 4.0; each episode's originating channel (RTV Business, ATN Bangla, and 11 episodes whose channel is recorded as unknown) is attributed in <code>registry.csv</code> . The <code>license_attestation</code> field in every registry row currently reads UNVERIFIED and must be completed before public release.
AI-generated / synthetic content	Disclosed. No synthetic data were generated, but all sentiment labels are AI-generated : 100% of the 1,811 turn labels and all 150 speaker verdicts were produced by Claude Opus 5 under a human-authored rubric, in a single pass. Human effort went into curation rather than labelling, covering rubric design, diarization repair, unit screening, out-of-domain exclusion and the discarding of stale blocks, all itemised with counts in Section 5.5. No human annotator re-labelled individual turns , and the annotation drafts in the roll folders still record <code>human_verified: false</code> for all 5,395 prepared utterances. Transcripts are machine-generated by Whisper-medium and speaker segments by <code>pyannote</code> . Every score in this report is agreement with an AI judge, not with human truth.

14. Data and Code Availability

Item	Details
Repository name	GitHub: <code>mayer-doa-coder/bangla-financial-talkshow-sentiment</code> ; Zenodo is the intended archival host once a DOI is minted
Dataset DOI / persistent identifier	Not yet minted
Direct dataset URL	https://github.com/mayer-doa-coder/bangla-financial-talkshow-sentiment
Access instructions	Restricted to the course instructor at submission; intended to become public. Derived annotations and transcripts are released; source broadcast audio is not redistributed and is obtainable from the channel references in <code>registry.csv</code> .
Dataset license	CC BY 4.0 for the derived annotations, transcripts and timings <i>only</i> . Source broadcast audio is third-party copyrighted and is not covered by this licence and not redistributed (Section 13).
Code / notebook URL	Same repository, under <code>speaker_sentiment/</code> . Contains <code>build_speaker_corpus.py</code> , <code>llm_judge.py</code> , <code>make_datasets.py</code> , <code>predict_speakers.py</code> , <code>tfidf_baseline.py</code> and <code>Speaker_Sentiment_BanglaBERT.ipynb</code> .

Item	Details
Code version / commit / release	v1.0, build 2 (45-episode corpus); the archive is not under version control, so the release is identified by the dataset version
README & environment	speaker_sentiment/README.md (pipeline, caveats, exact run sequence); speaker_sentiment/requirements.txt; torch 2.12, transformers, scikit-learn, pandas, matplotlib

15. Student Contributions (CRedit-style)

Report contributions transparently. Each student should have a clear role. Typical roles include conceptualization, methodology, data curation, investigation, software, validation, formal analysis, visualization, writing, and project administration.

Two figures are reported per student because they answer different questions. **Data share** is measured, not negotiated: it is the mean of each student's share of episodes, speaker turns and transcribed words in the released corpus, computed directly from `turns.jsonl` and `speakers.jsonl`, and it is reproducible from the files in the archive. It is uneven because episode availability was uneven, and some programmes published far more usable broadcasts than others. **Overall share** is the team's agreed division of the whole workload, which includes the pipeline engineering, annotation-draft preparation, diarization repair, modelling, analysis and report writing that the data share cannot see. Students with smaller collections carried correspondingly more of that shared work, which is why the overall column is close to even; the role column below records which parts each student took. The evidence column is the machine-verifiable part of the claim.

Roll	Student	Data share (%)	Overall (%)	CRedit roles and what they covered	Evidence / files owned
2107001	Nafiz Ahmed	9.4	16.0	<i>Data curation; methodology; validation; investigation; writing (review and editing).</i> Collection and pipeline run for roll 2107001; audio hygiene and the 16 kHz mono preprocessing tier; weak-supervision and GER trial branches; co-author of the six genre-specific rubric rules and reviewer of the rubric's borderline cases	2107001/results/: 4 episodes, 2.7 h, 239 turns, 20,670 words, 12 judged units, 184 turn labels; the only bundle carrying a complete processed/, weak/ and ger/ tier (37 files, 714 MB)
2107004	Tawhidul Hasan	21.7	17.0	<i>Software; data curation; conceptualization; project administration; writing (review and editing).</i> Collection and pipeline run for roll 2107004; the corpus-build, judge-merge and split scripts; cross-roll consolidation of six differing folder layouts into one schema	2107004/results/: 10 episodes, 5.6 h, 570 turns, 42,622 words, 29 judged units, 425 turn labels; CONSOLIDATION.md, aggregated/ STATUS.md, 10 annotation drafts, weak_labels/ (84 files)

Roll	Student	Data share (%)	Over-all (%)	CRedit roles and what they covered	Evidence / files owned
2107006	Sarwad Hasan Siddiqui	30.1	17.5	<i>Conceptualization; methodology; formal analysis; writing (original draft); project administration; data curation.</i> Largest collection share; annotation rubric and the out-of-domain exclusion criteria; report preparation; corresponding student	2107006/results/: 14 episodes, 9.4 h, 625 turns, 73,091 words, 60 judged units, 459 turn labels; 8 aggregated/quality-gate files (143 files, 828 MB)
2107009	Saif Ahmed Shejan	9.8	16.0	<i>Validation; data curation; investigation; writing (review and editing).</i> Collection and pipeline run for roll 2107009; the diarization quality audit that identified build 1's worst defect and the pyannotate re-run that repaired it, turning 263 unusable fragments on ep001 into 47 real speaker turns	2107009/results/: 5 episodes, 2.8 h, 190 turns, 22,498 words, 15 judged units, 151 turn labels; 15 diarization artefacts over 5 episodes (three passes per episode)
2107010	Maruf Shafiq	10.0	16.0	<i>Visualization; formal analysis; data curation; writing (original draft) (results); investigation.</i> Collection and pipeline run for roll 2107010; the confusion matrix, per-class and role/transcript-quality breakdowns; the qualitative error analysis behind Section 11	2107010/Results/: 4 episodes, 2.7 h, 289 turns, 20,131 words, 11 judged units, 205 turn labels; 18 diarization artefacts over 4 episodes; annotated/, weak_labels/
2107015	Dewan Salman Rahman Zisan	19.0	17.5	<i>Software; formal analysis; visualization; data curation; writing (review and editing).</i> Collection and pipeline run for roll 2107015; BanglaBERT fine-tuning, the aggregation-rule study and the hierarchical speaker model; the corpus diversity audit	2107015/results/: 8 episodes, 5.3 h, 501 turns, 40,264 words, 23 judged units, 387 turn labels; models/, artifacts/, reports/diversity_audit.md and 5 report figures

16. Acknowledgements, Funding and Competing Interests

16.1 Acknowledgements / Funding

We acknowledge the course instructors of CSE 4112 for guidance, and the authors of `csebuetnlp/banglabert`, `pyannotate.audio` and the BengaliAI Whisper-medium ASR model, whose openly released models made this pipeline possible.

This work received no external funding.

16.2 Declaration of Competing Interests

The authors declare no competing interests relevant to this dataset report.

17. References

Use numbered references in order of citation. Include sources for the problem domain, data-collection/measurement method, data source(s), instruments/software, related datasets, and any ML baseline methods that are actually used. A concise list (typically ≤ 20 references) is sufficient for a data-focused report.

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Appendix A. Complete Data Dictionary (if needed)

Place the full variable/field dictionary here if Section 7.2 would otherwise become too long.

A.1 turns.jsonl: 2,414 rows, 19 fields

Field	Type	Definition
turn_id	string	roll::episode::turnNNNN, globally unique
roll	string	Collecting student's roll number
episode	string	Episode identifier local to the roll (epNNN)
global_episode	string	roll::episode; the grouping key for all splits
turn_index	integer	Ordinal position of the turn within the episode
speaker_id	integer	Diarization cluster index, episode-local; -1 = unassigned
start_sec, end_sec, dur_sec	float	Turn boundaries and duration, seconds
text	string	Normalised Bangla transcript
text_raw	string	Transcript before normalisation
n_words	integer	Whitespace word count
n_windows	integer	Number of ASR decode windows merged into this turn
asr_confidence	float	Mean log-probability over the turn's words (≤ 0)
diarization_ provider	string	pyannotate or fallback
diarization_ trusted	boolean	True for all 45 episodes in this build
flag_ repetition_loop	boolean	ASR degenerate-repetition failure detected
flag_too_short	boolean	Turn below the minimum usable length
role	string	host / guest / minor / unassigned

A.2 speakers.jsonl: 212 rows. Adds speaker_key, programme, n_turns, n_usable_turns, n_words, n_clean_words, dur_sec, mean_words_per_turn, n_loop_turns, mean_asr_confidence, script (concatenated turns), turn_ids and judgeable.

A.3 judgements.jsonl: 150 rows. pass_id, judge_model, speaker_label, stance_score, speaker_confidence, speaker_summary, speaker_rationale, dominant_topics, transcript_quality, n_turns_sent, n_turns_returned, turn_labels (list of {turn_id, label, confidence, evidence}) and usage.

A.4 speakers_labelled.jsonl: 150 rows. Adds speaker_level_valid, per-class counts, net_polarity, net_polarity_word_weighted, the five aggregation outputs (agg_majority_vote, agg_net_polarity, agg_word_weighted, agg_conf_weighted, agg_word_conf_weighted) and split.

A.5 turns_{train,val,test}.jsonl: 1,009 / 297 / 324 rows. Turn fields plus speaker_key, label, judge_confidence, evidence, n_passes, pass_agreement, weight and split.

A.6 registry.csv: 32 rows. episode_id, channel, duration_sec, programme, has_remote_guest, covers_market_news, local_path, sha256, license_attestation, sample_rate, channels, codec, bit_rate_kbps, file_size_bytes, duration_flag, rejection_reason, sha256_verified, notes.

Appendix B. Optional Supporting Material

- Representative raw samples: an excerpt of a fused episode JSON showing word-level timings, probabilities and speaker assignment, alongside the reconstructed turn.
- Screenshots of the retrieval demonstration application (demo/app.py), which indexes every roll bundle and exposes speaker-filtered exact, fuzzy, semantic and hybrid search over the corpus with playable audio excerpts.

- Labelling guideline: the full judge rubric with worked annotation examples for each of the six genre-specific rules.
- Additional validation tables: the aggregation-rule agreement study (five rules, bottom-up vs. top-down), the subword length distribution used to choose `max_len`, and the per-episode diarization-provider breakdown.
- Environment specification and reproducibility notes: model revision hashes, seeds (pipeline 17, modelling 7), and measured CPU/GPU timings.
- The financial-target taxonomy (52 canonical targets with stems and orientation) and the 103-entry polarity-cue lexicon.

Template basis: Data in Brief data-article structure; Scientific Data Data Descriptor structure; MDPI Data Data Descriptor structure; plus CSE 4112-specific ML workflow and leakage-control requirements.

Appendix C. Dataset Release and FAIR-Readiness Checklist

Checklist item	Criterion	Yes/No	Evidence / note
Findable	Repository and persistent identifier/URL are provided; file/folder names are meaningful; metadata are searchable.	☒ Yes / ☐ No	GitHub: mayer-doa-coder/ bangla-financial-talkshow-sentiment DOI at archival release.
Accessible	Access instructions work; restrictions are justified; files can be downloaded by the intended user.	☒ Yes / ☐ No	Audio not redistributed; registry.csv enables independent retrieval
Interoperable	Common/open formats and standard units/labels are used where possible; schemas/codings are documented.	☒ Yes / ☐ No	JSON / JSONL / CSV / WAV / RTTM; fused schema 2.0; Appendix A
Reusable	License is explicit; README, data dictionary, provenance, limitations and citation information are provided.	☒ Yes / ☐ No	CC BY 4.0 (derived annotations); README, Appendix A, Sections 9 and 12 present
Raw + processed separation	Original observations are preserved separately from cleaned/derived data.	☒ Yes / ☐ No	Raw audio kept unmodified locally and excluded from the repository (Section 13); derived artefacts under results/
Versioning	Dataset version/date and change history are recorded.	☒ Yes / ☐ No	v1.0, 15 September 2026; build 1 → build 2 changes recorded in Section 4
Integrity	Duplicate/corruption/schema checks are complete; checksums are included where useful.	☒ Yes / ☐ No	SHA-256 in registry.csv and run-manifest.json; transcript-hash dedup
Leakage safety	Related samples/subjects/augmentations do not cross ML data partitions.	☒ Yes / ☐ No	Episode-disjoint split asserted in code; programme-level overlap remains (Section 12)

Checklist item	Criterion	Yes/No	Evidence / note
Reproducible processing	Code/notebook can reproduce processed data and baseline pipeline from the released inputs.	☒ Yes / ☐ No	build_speaker_corpus.py + make_datasets.py + notebook; seeds 17 / 7
Ethics/privacy	Consent/approval/licensing/anonymization requirements are satisfied.	☐ Yes / ☒ No	Episode-local speaker indices only, but license_attestation is still UNVERIFIED for all 56 rows
Documentation	README explains files, variables, setup and reuse; captions/tables match the repository contents.	☒ Yes / ☐ No	speaker_sentiment / README.md
Team accountability	All six students' contributions are recorded.	☒ Yes / ☐ No	Section 15